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Test Cases	Test Data	Expected Result	Test Result	Pass/Fail
D1 - valid driverID passes	driverID = "29@!aaaaAB"	validateDriverIdFormat returns normally (no exception thrown)	No exception thrown	Pass
D1 - driverID with wrong length is rejected	driverID = "29@!aaaaA" (9 chars) and "29@!aaaaABC" (11 chars)	ValidationException thrown with message about 10-character length	ValidationException thrown in both cases	Pass
D1 - first two characters of driverID must be digits 2-9	driverID = "19@!aaaaAB" and "09@!aaaaAB"	ValidationException thrown because first digits must be 2-9	ValidationException thrown	Pass
D1 - at least 2 special characters between positions 3-8	driverID = "23#bbbbbbAB" (1 special, fail), "23#\$bbbbAB" (2 specials, pass)	1 special -> ValidationException; 2 specials -> no exception	Behaviour matches expected boundary	Pass
D2 - valid 5-part pipe-separated address passes	address = "10 Smith St Sydney NSW Australia"	validateAddressFormat returns normally	No exception thrown	Pass
D2 - missing component or empty trailing field is rejected	address = "10 Smith St Sydney NSW" and "10 Smith St Sydney NSW "	ValidationException thrown for both	ValidationException thrown	Pass
D2 - extra component beyond 5 parts is rejected	address = "10 Smith St Sydney NSW Australia Extra"	ValidationException thrown (must be exactly 5 parts)	ValidationException thrown	Pass
D3 - valid DD-MM-YYYY birthdate is accepted	birthdate = "12-05-1990"	validateBirthdateFormat returns normally	No exception thrown	Pass
D3 - wrong format (YYYY-MM-DD or slashes) is rejected	birthdate = "1990-05-12" and "12/05/1990"	ValidationException thrown for both	ValidationException thrown	Pass
D3 - future birthdate is	birthdate = "12-05-2999"	ValidationException thrown because	ValidationException thrown	Pass

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rejected (edge case)		birthdate cannot be in the future		
D4 - driver with <= 10 years experience may change licenseType	existing experience = 3, license "Light" -> updated license "Medium"	validateForUpdate returns normally	No exception thrown	Pass
D4 - driver with 12 years experience cannot change licenseType	existing experience = 12, license "Heavy" -> updated license "PublicTransport"	ValidationException thrown (D4)	ValidationException thrown	Pass
D4 - boundary: exactly 10 years experience may still change license	existing experience = 10, license "Light" -> updated license "Heavy"	validateForUpdate returns normally ("more than 10" -> strict >)	No exception thrown	Pass
D4 - boundary: 11 years experience cannot change license	existing experience = 11, license "Light" -> updated license "Heavy"	ValidationException thrown (just over the threshold)	ValidationException thrown	Pass
D5 - updating only address and birthdate succeeds	driverID and name unchanged, only address/birthdate differ	validateForUpdate returns normally	No exception thrown	Pass
D5 - changing driverID is rejected	updated driverID differs from existing driverID	ValidationException thrown (D5)	ValidationException thrown	Pass
D5 - changing name is rejected	updated name differs from existing name	ValidationException thrown (D5)	ValidationException thrown	Pass
D5 - no-op update with identical fields succeeds (edge)	all fields equal between existing and updated	validateForUpdate returns normally (unchanged is allowed)	No exception thrown	Pass
Composite - validateForAdd accepts a fully-valid driver	Valid Driver { 29@!aaaaAB, Alice, 3y, Light, valid addr, valid dob }	validateForAdd returns normally	No exception thrown	Pass

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D1 unique - duplicate driverID is rejected via validateForAdd	existingIDs contains "29@!aaaaAB"; adding driver with same ID	ValidationException with "already exists"	ValidationException thrown	Pass
B1 - valid 8-digit busID passes	busID = "12345678"	validateBusIdFormat returns normally	No exception thrown	Pass
B1 - 7 or 9 digit busID is rejected	busID = "1234567" and "123456789"	ValidationException thrown for both	ValidationException thrown	Pass
B1 - busID containing non-digit characters is rejected	busID = "1234A678" and "1234-678"	ValidationException thrown for both	ValidationException thrown	Pass
B1 - duplicate busID is rejected (uniqueness)	existingIDs contains "12345678"; checking same ID	ValidationException thrown with "B1" message	ValidationException thrown	Pass
B2 - decreasing capacity passes	existing capacity = 50 -> updated capacity = 40	validateForUpdate returns normally	No exception thrown	Pass
B2 - increasing capacity is rejected	existing capacity = 40 -> updated capacity = 50	ValidationException thrown (B2)	ValidationException thrown	Pass
B2 - equal capacity passes (edge boundary)	existing capacity = 40 -> updated capacity = 40	validateForUpdate returns normally	No exception thrown	Pass
B3 - young driver may drive large bus	driver age = 30, capacity = 60	validateAssignment returns normally	No exception thrown	Pass
B3 - 60-year-old driver cannot drive bus with capacity 50	driver age = 60, capacity = 50	ValidationException thrown (B3)	ValidationException thrown	Pass
B3 - 51-year-old driver may drive small bus (capacity 49)	driver age = 51, capacity = 49	validateAssignment returns normally	No exception thrown	Pass

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B3 - boundary: exactly 50 years old, capacity 50 is allowed	driver age = 50, capacity = 50	validateAssignment returns normally ("older than 50" -> strict >)	No exception thrown	Pass
B4 - experienced driver (6y) on electric bus succeeds	driver experience = 6, fuelType = Electricity, license Heavy	validateAssignment returns normally	No exception thrown	Pass
B4 - driver with 3 years experience on electric bus is rejected	driver experience = 3, fuelType = Electricity, license Heavy	ValidationException thrown (B4)	ValidationException thrown	Pass
B4 - boundary: exactly 5 years experience is allowed for electric	driver experience = 5, fuelType = Electricity, license Heavy	validateAssignment returns normally	No exception thrown	Pass
B4 - inexperienced driver on diesel bus is still allowed	driver experience = 1, fuelType = Diesel	validateAssignment returns normally (B4 applies only to electric)	No exception thrown	Pass
B5 - Heavy licence on Hybrid bus is allowed	driver license = Heavy, fuelType = Hybrid	validateAssignment returns normally	No exception thrown	Pass
B5 - Light licence on Electricity bus is rejected	driver license = Light, fuelType = Electricity (experience 8y)	ValidationException thrown (B5)	ValidationException thrown	Pass
B5 - PublicTransport licence on Diesel bus is unrestricted (edge)	driver license = PublicTransport, fuelType = Diesel	validateAssignment returns normally	No exception thrown	Pass
Composite - validateForAdd accepts a fully- valid bus	Valid Bus { 12345678, 40, 50.0, Diesel }	validateForAdd returns normally	No exception thrown	Pass

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Validation - negative capacity is rejected on add	Bus { 12345678, -5, 50.0, Diesel }	ValidationException about positive capacity	ValidationException thrown	Pass
INTEGRATION - valid drivers are persisted to drivers.txt	Add Alice (29@!aaaaAB) and Bob (37#\$bbbbCD) via repository	TXT file contains both records; reloaded repo finds both	TXT file contents and reloaded count = 2 verified	Pass
INTEGRATION - invalid drivers are rejected and not written	Add valid Alice; then attempt duplicate, bad address, bad birthdate	All invalid adds throw ValidationException; count remains 1	count() == 1; file does not contain rejected records	Pass
INTEGRATION - update is persisted; reload sees new license/address	Add Alice, update license Light->Heavy and address	Fresh repo on same file reads updated license and address	Reloaded driver has license Heavy and updated address	Pass
INTEGRATION - count() is accurate after add/reject/reload	Add 2 valid; attempt duplicate (rejected); reload	Count = 2 after adds, 2 after reject, 2 after reload	count() == 2 in all three checks	Pass
INTEGRATION - D4 license lock enforced through repository.update	Add driver with 12y experience; attempt licenseType change	update throws ValidationException; file unchanged	License remains Heavy after failed update	Pass
INTEGRATION - valid buses are persisted to buses.txt	Add Bus 12345678 (Diesel) and 87654321 (Hybrid)	TXT file contains both records; reloaded repo finds both	Reloaded count = 2; fuelType + capacity match	Pass
INTEGRATION - invalid buses are rejected and not written	Add valid; attempt duplicate, 7-digit, non-digit, bad fuelType	All invalid adds throw ValidationException; count remains 1	count() == 1; rejected records absent from file	Pass
INTEGRATION - capacity decrease update is persisted; increase fails	Update Bus 12345678 from capacity 50 to 35, then try 35 -> 60	First update succeeds and persists; second update throws	File shows capacity 35; second attempt throws ValidationException	Pass
INTEGRATION - bus count is accurate after add/reject/reload	Add 3 valid buses; attempt duplicate ID add	Count = 3 after adds, stays 3 after rejected add, 3 after reload	count() == 3 in all three checks	Pass

INTEGRATION - assignDriver enforces B3/B4/B5 at repository level	Assign junior (B4 fail), Light-license (B5 fail), then valid Heavy 10y	First two throw ValidationException; third succeeds and persists	Assigned driver ID written to TXT file and read back	Pass
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Description

Test Cases: Description of what is being tested

Test Data: Input values used for the test

Expected Result: The expected system behaviour/output

Test Result: The actual outcome observed during testing

Pass/Fail: Whether the test case passed or failed based on the comparison of expected and actual results